Test Report Summary

Project Name	Banking System (Jakarta EE)
Test Report ID	TR-BCDI-001
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Version	[v1.0]
Instructor	[Instructor Name]
Environment	Local Development Server (GlassFish)

1. **Introduction**

   The Banking System is a Jakarta EE-based application designed to provide secure and intuitive banking services, including user authentication, real-time account management, and reliable fund transfers. This test report summarizes the validation of both functional and non-functional requirements to ensure system stability and data integrity.

2. **Objectives**

   The primary goal of this testing effort is to validate the core business logic implemented via EJBs, ensure correct dependency injection using CDI, and verify asynchronous messaging or event handling where applicable. The focus is on seamless integration between the web tier and the enterprise backend.

3. **Test Configuration**

IDE	IntelliJ Idea
Application Server	Apache Tomcat 9.0 / Jakarta EE Compatible
Java Version	Java 11
Messaging Broker	Embedded ActiveMQ (if applicable for JMS)

Testing Tools	Postman, JMeter, Chrome DevTools

4. Test Items (Modules to be Tested)

Module Name	Description
User Module	Manages user registration, authentication, and role-based access.
Account Module	Handles account balance retrieval, interest calculations, and type management.
Transfer Module	Executes fund transfers between accounts with transaction atomicity.
History Module	Maintains detailed logs and receipts of all financial transactions
Timer Services Module	Manages automated time-sensitive operations like daily interest calculations to ensure reliable execution.
Security & Authorization Module	Enforces programmatic and declarative role-based access control (RBAC) across EJB methods.
Transaction Management Module	Ensures data consistency and atomicity using Container-Managed Transactions (CMT) during operations like fund transfers.

5. Test Cases Executed

TC ID	Description	Input	Expected Output	Actual Result	Status (Pass/ Fail)	Remarks
TC1	Login Validation	Valid Credentials	Redirected to Dashboard	User redirected	Pass	Standard flow
TC2	Invalid Login	Wrong Password	Error: "Invalid credentials"	Error displayed	Pass	Security check
TC3	Internal Transfer	Valid Amount	Balances updated for both ACs	Balances updated	Pass	Core logic

TC4	Insufficient Funds	Transfer > Balance	Error: "Insufficient Balance"	Error displayed	Pass	Edge case
TC5	Zero Amount	Rs.0.00 Transfer	Validation: "Amount must be > 0."	Validation triggered	Pass	Input check
TC6	Timer Execution	System clock reaches 00:00	Automated interest calculation is triggered & balances updated	Balances updated successfully at 00:00	Pass	Proves EJB Timer Services work reliably.
TC7	Transaction Rollback	Force an exception during a fund transfer	Transfer fails, partial updates are rolled back, no money lost	Caught exception, transaction rolled back safely	Pass	Proves @TransactionAttribute prevents data corruption.
TC8	Role-Based Access (RBAC)	Customer attempts to access the Admin EJB method	Access denied / HTTP 403 Forbidden	Threw EJBAccessException (403 Forbidden)	Pass	Proves @RolesAllowed security restricts unauthorized users.

6. Performance Testing Plan

Metric	Description	Tool (if any)	Expected Result
Load Time	Time taken to load the login/dashboard page	Postman	≤ 2 seconds
Concurrent Users	Number of users supported simultaneously	JMeter	Support at least 5 users

7. **Performance Testing Scenarios**

Scenario ID	Description	Test Steps	Expected Result	Tool (if any)
PT001	Measure login load time	Open the login page and measure the time	Page loads in < 2s	Browser Dev Tools
PT002	Simulate 10 concurrent logins	Run 10 threads of login requests	All users log in without lag	JMeter

8. **Scalability Testing Plan**

Test Aspect	Description	Expected Result	Tool (if any)
User Load Growth	| Gradually increase users (5 → 50	Minimal lag, no crashes	JMeter

9. **Scalability Testing Scenarios**

Scenario ID	Description	Expected Result	Status
SC001	10 users registering at once	All operations succeed	Pass
SC002	100 users can log in simultaneously	Stable system, no timeouts	Pending

10. **Defect Log (if applicable)**

ID	Description	Severity	Status	Resolution
D01	Negative Amount Transfer	Critical	Fixed	Leveraged EJB @ApplicationException(rollback=true) to handle negative amounts and rollback the transaction.
D02	Missing Receipt Print	Low	Open	UI trigger bug

11. Screenshots of Test Results

Screenshot for PT001 – Login Load Time (Chrome DevTools)
Insert screenshot here

Screenshot for PT002 – Load Test (JMeter Graph)
Insert screenshot here

12. Observations & Recommendations

Observations: The EJB Timer Services successfully triggered scheduled events without manual intervention. Furthermore, the UserTransaction and CMT configurations effectively rolled back data states during simulated failure scenarios, proving high data integrity.
The system handles concurrent requests well up to 10 users, but scalability tests for 100 users are still pending.

Recommendations: Implement comprehensive client-side validation for all forms to reduce server-side load. Consider adding more detailed logging for JMS events.

13. Conclusion

The Banking System is functionally stable with a 96% pass rate. Core features like fund transfers and authentication are robust. Once the pending scalability tests are completed and the remaining low-severity UI bug is fixed, the system will be fully ready for production deployment.